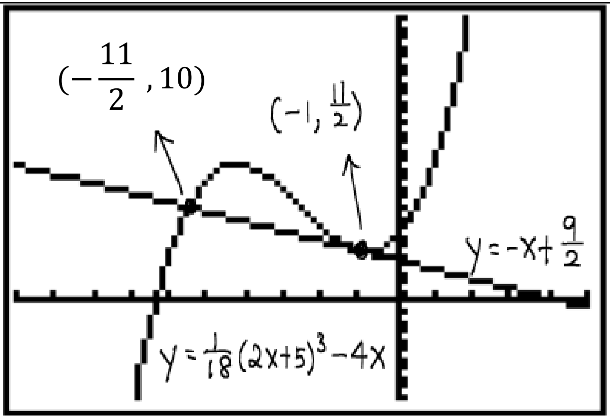
RVHS 2013 Year 6 Preliminary Examination: 8864 H1 Mathematics Solution

	Question 1 [5 marks]	
1	P = Sales – Costs $= CN - (250 + 0.9N)$ $= C(600 - 200C) - (250 + 0.9(600 - 200C))$ $= 600C - 200C^2 - (250 + 540 - 180C)$ $= -200C^2 + 780C - 790$	
	Consider discriminant. $b^2 - 4ac = 780^2 - 4(-200)(-790) = -23600 < 0$ Since discriminant < 0, and coefficient of $C^2 < 0$, Profit < 0 for all C. It will not be profitable for River Valley Bakery to sell hurry puffs because they will not be able to make enough money to cover the cost of sales, no matter what price they sell puffs at.	

	Question 2 [7 marks]	
(i)	$\frac{dy}{dx} = \frac{1}{3}(2x+5)^2 - 4$ $y = \int \frac{1}{3}(2x+5)^2 - 4 \ dx$ $= \frac{1}{18}(2x+5)^3 - 4x + c$ Substitute $\left(2, 32\frac{1}{2}\right)$: $\frac{65}{2} = \frac{1}{18}(9)^3 - 4(2) + c \quad \Rightarrow \quad c = 0$ $\therefore y = \frac{1}{18}(2x+5)^3 - 4x$ (Ans)	
(ii)	When $x = -1$, $\frac{dy}{dx} = \frac{1}{3}(3)^2 - 4 = -1; \quad y = \frac{1}{18}(3)^3 - 4(-1) = \frac{11}{2}$ Equation of tangent: $y - \frac{11}{2} = -1(x+1)$ $\therefore y = -x + \frac{9}{2}$ (Ans)	



$$\frac{1}{18}(2x+5)^3 - 3x - \frac{9}{2} > 0$$

$$\frac{1}{18}(2x+5)^3 - 4x > -x + \frac{9}{2}$$

Using GC, we see that the tangent meets curve C again at the points $\left(-\frac{11}{2},\ 10\right)$ and $\left(-1,\ \frac{11}{2}\right)$.

Thus, we are finding the range of values of *x* such that the curve *C* is above the tangent.

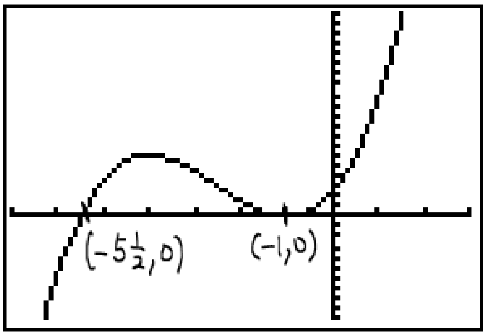
$\therefore$ Range of values of *x* is $x > -\frac{11}{2}, x \neq -1$ OR

$$x \in \left(-\frac{11}{2}, -1\right) \cup (-1, \infty)$$

<u>Otherwise method:</u>

Using GC, sketch the curve

$$y = \frac{1}{18}(2x+5)^3 - 3x - \frac{9}{2}$$

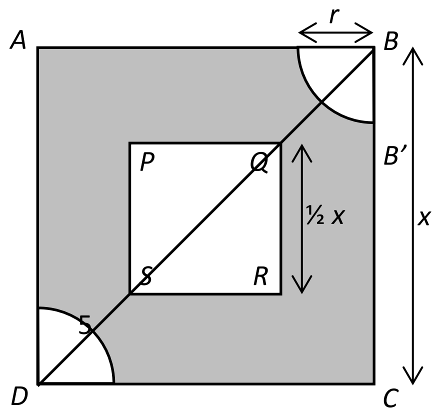


The curve cuts the *x*-axis at $\left(-\frac{11}{2},\ 0\right)$ and $(-1, 0)$.

$\therefore$ Range of values of *x* is $x > -\frac{11}{2}, x \neq -1$ OR

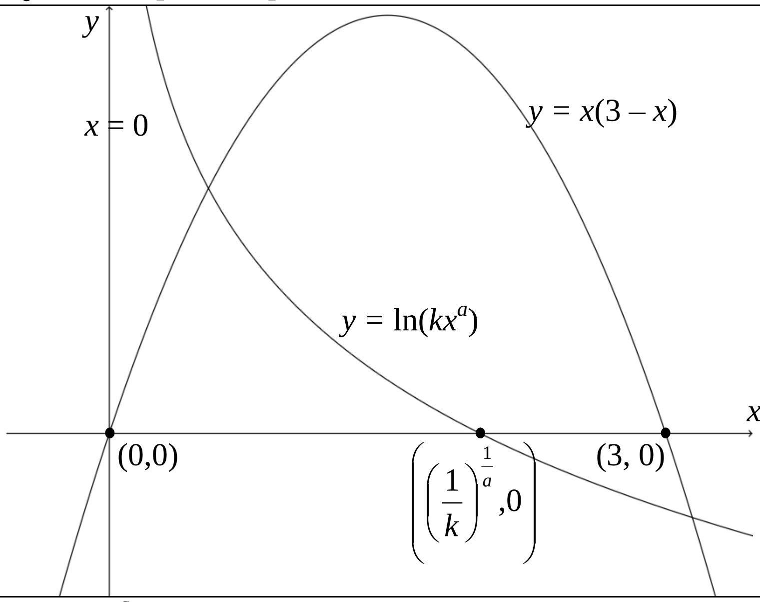
$$x \in \left(-\frac{11}{2}, -1\right) \cup (-1, \infty)$$

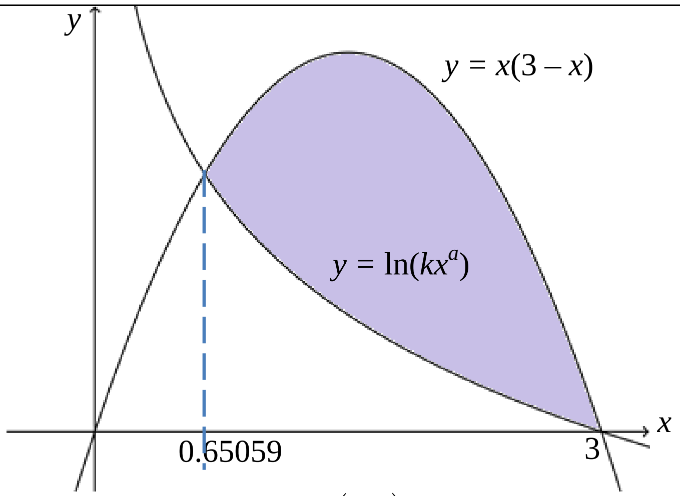
	Question 3 [7 marks]	
(i)	Let $\mathrm{f}(x)=\mathrm{e}^{x\ln x}$ and $\mathrm{f}'(x)=x^x\left(\ln x+1\right)$ $\int_1^2 \frac{x^x\left(\ln x+1\right)}{2\sqrt{\mathrm{e}^{x\ln x}}}\,\mathrm{d}x = \int_1^2 \frac{\mathrm{f}'(x)}{2\sqrt{\mathrm{f}(x)}}\,\mathrm{d}x$ $=\left[\sqrt{\mathrm{f}(x)}\right]_1^2$ $=\left[\sqrt{\mathrm{e}^{x\ln x}}\right]_1^2$ $=\sqrt{\mathrm{e}^{2\ln 2}}-\sqrt{\mathrm{e}^{1\ln 1}}$ $=\sqrt{4}-\sqrt{1}$ $=1$	
(ii)	$\int_0^a \frac{2x+1}{x^2+x+1}\,\mathrm{d}x = 2\int_1^2 \frac{x^x\left(\ln x+1\right)}{2\sqrt{\mathrm{e}^{x\ln x}}}\,\mathrm{d}x$ $\int_0^a \frac{2x+1}{x^2+x+1}\,\mathrm{d}x = 2$ $\left[\ln\left(x^2+x+1\right)\right]_0^a = 2$ $\left[\ln\left(a^2+a+1\right)\right]-\ln 1 = 2$ $a^2+a+1=\mathrm{e}^2$ $a^2+a+(1-\mathrm{e}^2)=0$ From GC, $a=2.0766$ or $a=-3.0766$ (rej $\because a>0$) $\therefore\ a=2.08$	

	Question 4 [7 marks]	
(i)	Since the drop takes 10 seconds to move from B to Q, So distance $BQ = 0.5\times 10 = 5\,\text{cm}$ Similarly, distance DS = 5 cm 	

	<u>Method 1:</u> By Pythagoras' Theorem, $5^2 = \left(\frac{1}{4}x\right)^2 + \left(\frac{1}{4}x\right)^2 = \frac{1}{8}x^2$ $25 = \frac{1}{8}x^2$ $x^2 = 200$ $x = \sqrt{200} = 10\sqrt{2}$ (reject negative) (shown) <u>Method 2:</u> By Pythagoras' Theorem, $QS^2 = \left(\frac{1}{2}x\right)^2 + \left(\frac{1}{2}x\right)^2 = \frac{1}{2}x^2$ $QS = \frac{1}{\sqrt{2}}x - - - - - (1)$ Also, $BD^2 = x^2 + x^2 = 2x^2$ $(BQ + QS + SD)^2 = 2x^2$ $(QS + 10)^2 = 2x^2$ $QS = \sqrt{2}x - 10 - - - - - (2)$ Equating (1) and (2), $\frac{1}{\sqrt{2}}x = \sqrt{2}x - 10$ $\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)x = 10$ $\frac{1}{\sqrt{2}}x = 10$ $\therefore x = 10\sqrt{2}$ (shown) --- Area $= x^2 - \left(\frac{1}{2}x\right)^2 - \frac{1}{2}\pi r^2$ $= \frac{3}{4}x^2 - \frac{1}{2}\pi r^2 = \frac{3}{4}\left(10\sqrt{2}\right)^2 - \frac{1}{2}\pi r^2$ $= 150 - \frac{1}{2}\pi r^2$	
(ii)	$\frac{dr}{dt} = 0.5$ Let S be the area of the cardboard which is dry.	

	$S=150-\frac{1}{2}\pi r^2 \Rightarrow \frac{dS}{dr}=-\pi r$ $\frac{dS}{dt}=\frac{dS}{dr}\cdot\frac{dr}{dt}$ When $r=2$, $\frac{dS}{dt}=-\pi(2)\cdot(0.5)=-\pi\ \text{cm}^2\text{s}^{-1}$ Thus S is decreasing at a rate of π cm^{2}s^{-1} at the end of the 4th second. Not suitable, because when r = 6, the expression $S=150-\frac{1}{2}\pi r^2$ does not hold, since $0\le r\le BQ=5$	

	Question 5 [9 marks]	
(i)		
(ii)	$y=\ln(kx^a)=\ln k+a\ln x$ When x = 3, y = (3)(3 – (3)) = 0 So, 0 = ln$k$ + $a$ln(3) ----- (1) When x = 0.65059, y = (0.65069)(3 – (0.65059)) = 1.52850265 So, 1.528502652 = ln$k$ + $a$ln(0.65059) ----- (2) Solving (1) and (2) simultaneously for ln$k$ and a, $\ln k=1.098620968 \Rightarrow k=3$ $a=-1$	

(iii)	 Area $= \int_{0.65069}^{3} x(3-x)-\ln\left(kx^{a}\right)\mathrm{d}x$ $= \int_{0.65069}^{3} x(3-x)-\ln(3x^{-1})\ \mathrm{d}x$ $= 2.601899087$ (from GC) ≈ 2.602 units^2	

Question 6 [4 Marks]		
(i)	Let X be the r.v. that denotes the servicing time of a random air-conditioning unit. Since n is large, by Central Limit Theorem, $\overline{X} \sim N\left(40, \dfrac{25^2}{n}\right)$ approximately $P(\overline{X} \le 45) \ge 0.95$ $P\left(Z \le \dfrac{45-40}{25/\sqrt{n}}\right) \ge 0.95$ $\dfrac{45-40}{25/\sqrt{n}} \ge 1.644853626$ $\dfrac{5}{25}\sqrt{n} \ge 1.644853626$ $\sqrt{n} \ge 8.22426813$ $n \ge 67.63858627$ Least $n = 68$	
(ii)	Assume the times taken to service the air-conditioning units are independent.	

	Question 7 [5 marks]	
(i)	A random sample here means the sample is obtained by selecting 100 students from the 2000 students in such a way that each of the 2000 students will have an equal probability of $\frac{100}{2000} = \frac{1}{20}$ of being selected to do the survey.	
(ii)	<table><tr><td>Calculator Company</td><td>A</td><td>B</td><td>C</td></tr><tr><td>Number of students to be surveyed</td><td>$\frac{520}{2000} \times 100$ $= 26$</td><td>$\frac{620}{2000} \times 100$ $= 31$</td><td>$\frac{860}{2000} \times 100$ $= 43$</td></tr></table> 26 students who use calculator A, 31 students who use calculator B and 43 students who use calculator C will be randomly selected to be surveyed.	
(iii)	Stratified sampling would ensure that each group (stratum) in the population is represented; while random sampling may have a chance of missing out an important group completely, or may end up with a certain group overly represented in the sample. Instead of the mentioned 3 strata, a better way could be to further divide the students in each stratum by gender or by level. (e.g. Company A, Sec 1; Company A, Sec 2; Company A, Sec 3 etc)	

	Question 8 [7 marks]	
(i)	For A and B to be mutually exclusive, $\mathrm{P}(A \cap B) = 0$ or $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$ Given $\mathrm{P}(A' \cup B') = \frac{9}{10}$, $\frac{9}{10} = \mathrm{P}(A' \cup B') = \mathrm{P}[(A \cap B)']$ $= 1 - \mathrm{P}(A \cap B)$ $\mathrm{P}(A \cap B) = \frac{1}{10} \neq 0$ $\therefore$ Events A and B are not mutually exclusive.	
(ii) (a)	$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$ $= \frac{3}{5} + \frac{1}{3} - \frac{1}{10}$ $= \frac{5}{6}$	

(b)	$\mathrm{P}(A \mid A\cup B)=\dfrac{\mathrm{P}[A\cap(A\cup B)]}{\mathrm{P}(A\cup B)}$ $=\dfrac{\mathrm{P}(A)}{\mathrm{P}(A\cup B)}$ $=\dfrac{3/5}{5/6}=\dfrac{18}{25}$	

	Question 9 [8 marks]	
(i)	Binomial distribution is suitable because (1) There is a fixed number of patients during the trial (no patients are added or removed once the trial begins). (2) Each patient has approximately the same chance of responding positively. (3) There are only two mutually exclusive outcomes, patients either respond positively or don't respond positively. Assume that the response of a patient to the potential cure is independent of the others'.	
(ii)	$X \sim B(200, p_1)$ where $p_1=\dfrac{p}{100}$ $\mathrm{Var}(X) = np_1(1-p_1)$ $32 = 200(p_1 - p_1^2)$ -----(1) $200p_1^2 - 200p_1 + 32 = 0$ $p_1 = 0.2$ or 0.8 (rejected because $p_1 < 0.5$) So $p = 20$ (shown)	
(iii)	$\mathrm{P}(X\geq 8 \mid X<10)=\dfrac{\mathrm{P}(X\geq 8\cap X\leq 9)}{\mathrm{P}(X\leq 9)}$ $=\dfrac{\mathrm{P}(8\leq \mathrm{X}\leq 9)}{\mathrm{P}(X\leq 9)}$ $=\dfrac{\mathrm{P}(X=8)+\mathrm{P}(X=9)}{\mathrm{P}(X\leq 9)}$ $= 0.9704267234 \approx 0.970$	
(iv)	Let Y be the r.v. denoting the number of patients who respond positively out of n. Then Y~B(n, 0.2). $\mathrm{P}(Y\geq 1)>0.95$ $1-\mathrm{P}(Y=0)>0.95$ $\mathrm{P}(Y=0)<0.05$ $0.8^n<0.05$	

Method 1: Taking logarithms

$n \ln 0.8 < \ln 0.05$

$$n > \frac{\ln 0.05}{\ln 0.8} = 13.4251$$

Method 2: Graphical Approach (1)

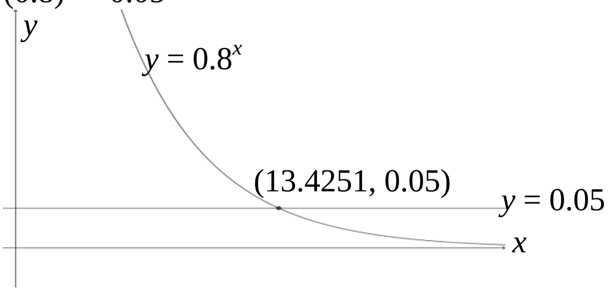
$P(Y = 0) < 0.05$

Plotting y = binompdf(X, 0.2,0) and from stat table below, least value of $n = 14$.

n	$P(Y = 0)$
13	0.054976
14	0.04398

Method 3: Graphical Approach (2)

$(0.8)^n < 0.05$



Therefore, the least value number of patients required for the trial would be 14.

Question 10 [10 marks]

(i) Unbiased estimate of the population mean,

$$\bar{x} = \frac{\sum x}{n} = \frac{825}{50} = 16.5$$

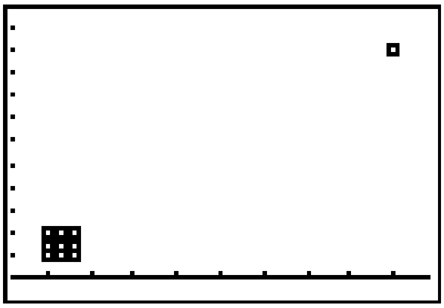
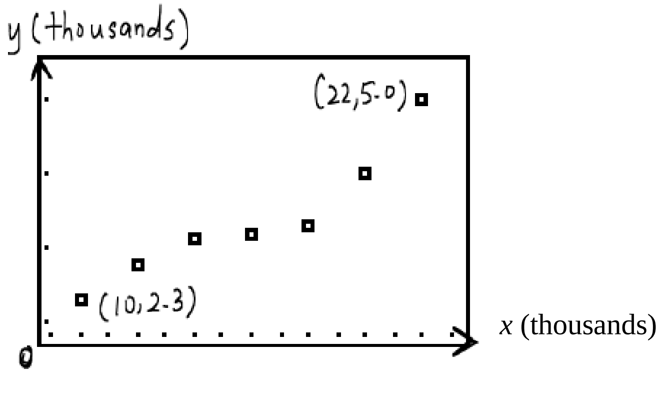
$\sum (x-16)^2 = 285$ (Given) and

$\sum (x-16) = \sum x - \sum 16 = 825 - 50(16) = 25$

$$s^2 = \frac{1}{n-1}\left[\sum (x-16)^2 - \frac{\left(\sum (x-16)\right)^2}{n}\right]$$

$$= \frac{1}{49}\left[285 - \frac{25^2}{50}\right] = 5.56122449$$

	Let μ be the mean time taken by an author to write a book. Test $H_0 : \mu = 16$ Against $H_1 : \mu > 16$ 1-tailed test at α % level of significance Under H_0, $\overline{X} \sim \mathrm{N}\left(16, \dfrac{5.56122449}{50}\right)$ approx. Test statistic: $Z = \dfrac{\overline{X} - 16}{\sqrt{\dfrac{5.56122449}{50}}} \sim$ N(0, 1) approx. From GC, p value = 0.0669063306 Since the claim is valid, we do not reject H_0, so $p \geq \dfrac{\alpha}{100}$ $\Rightarrow 0.0669063306 \geq \dfrac{\alpha}{100}$ $\therefore \alpha \leq 6.69063306$ Ans: $\alpha \leq 6.69$	
(ii)	The p-value of 0.0694 means that the smallest significance level for which the claim that the mean time spent on writing one textbook is (at most) 16 months would be rejected, is 6.94%	
(iiia)	Since H_0 is rejected, so p-value < 0.10. So p-value must also be less than 0.20 Thus, H_0 is rejected at 20% significance level. The company's claim is invalid.	
(iiib)	This is a 2-tailed test. For 2-tailed test, the p-value will be double that of the corresponding 1-tailed test. We only know that the p-value of the 1-tailed test is less than 0.10, but have no idea about its exact value. So the p-value of the 2-tailed test may or may not be less than 0.10. H_0 may or may not be rejected. That is, given only that piece of information, we cannot determine the conclusion of this new test.	

	Question 11 [13 marks]	
(a)	The scatter diagram gives a good overview of how the two variables are related. 	
(b)(i)	$y = 0.191x + 0.329$ $\Rightarrow \bar{y} = 0.191\bar{x} + 0.329$ since $(\bar{x}, \bar{y})$ lies on the regression line of y on x. $\bar{x} = \frac{10+12+14+16+18+20+22}{7} = 16$ $\bar{y} = 0.191(16) + 0.329 = 3.385$ $\bar{y} = \frac{2.3+2.8+3.1+k+3.3+4.0+5.0}{7} = \frac{20.5+k}{7}$ $\frac{20.5+k}{7} = 3.385$ $k = 3.195 = 3.2$ (1 decimal place) (shown) The figure 0.191 suggests that for every additional 1000 bottles of detergent sold, the profit increases by \$191.	
(ii)		
(iii)	Using GC, r = 0.9398991934 ≈ 0.940 This r value suggests a <u>strong positive linear correlation</u> between the number of bottles sold and the profit. However the scatter diagram indicates some degree of curvilinear relationship, hence the value of r alone is not sufficient.	

	If the number of bottles sold is given in dozens, the r should remain unchanged because the r value is not affected by any linear transformations of its random variables.	
(iv)	Using GC, the least squares regression line of x on y is $x = 4.62345679y + 0.3462962963$ i.e. $x = 4.62y + 0.346$	
(v)	Since number of bottles sold is an independent variable, we use the regression line of y on x. When y = 4.5 thousand dollars, $4.5 = 0.191x + 0.329$ $x = 21.83769634 \approx 21.8$ <u>thousands</u>	

Question 12 [13 marks]		
(i)	Let A and B be the mass, in milligrams, of a randomly chosen carton of apple juice and blueberry juice respectively. $A \sim \mathrm{N}\left(51,\ 3^2\right) \qquad B \sim \mathrm{N}\left(\mu,\ \sigma^2\right)$ Given P($B < 76$) = P($B > 160$) By symmetry, $\mu = \dfrac{76+160}{2} = 118$ (shown)	
(ii)	$\mathrm{P}(B < 115) = \dfrac{2}{5}\mathrm{P}(B < 121)$ $\mathrm{P}(B < 115) = \dfrac{2}{5}\left[1 - \mathrm{P}(B \geq 121)\right]$ $\mathrm{P}(B < 115) = \dfrac{2}{5}\left[1 - \mathrm{P}(B < 115)\right]$ $\dfrac{7}{5}\mathrm{P}(B < 115) = \dfrac{2}{5}$ $\mathrm{P}(B < 115) = \dfrac{2}{7}$ $\mathrm{P}\left(Z < \dfrac{115-118}{\sigma}\right) = \dfrac{2}{7}$ $\mathrm{P}\left(Z < \dfrac{-3}{\sigma}\right) = \dfrac{2}{7}$ By InvNorm, $-\frac{3}{\sigma} = -0.565948817$ $\sigma = 5.300832707 \approx 5.30$ (shown)	

(iii)	Let A be the mass, in milligrams, of a randomly chosen carton of apple juice. $A \sim \mathrm{N}\left(51,\ 3^2\right)$ $2\mathrm{P}\left(A > 51\right)\mathrm{P}\left(A < 51\right) = 2\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$ $= \frac{1}{2}$	
(iv)	$\left[\mathrm{P}(B > 115)\right]^5 = 0.7142857521^5$ $= 0.1859344813 \approx 0.186$ OR $\left[\mathrm{P}(B > 115)\right]^5 = \left(\frac{5}{7}\right)^5$ $= 0.1859344321 \approx 0.186$	
(v)	Let X be the r.v. for the number of cartons of apple juice with vitamin C of more than 54 mg, out of all 60 cartons $\mathrm{P}(A > 54) = 0.1586552596$ $X \sim \mathrm{B}\left(60, 0.1586552596\right)$ Since $n = 60 > 50$ is large, $np = 60 \times 0.1586552596 = 9.519315574 > 5$, and $nq = 60 \times \left(1 - 0.1586552596\right) = 50.48068443 > 5$ Hence $X \sim \mathrm{N}\left(9.519315574, 8.009026091\right)$ approx $\mathrm{P}\left(X > 15\right) \overset{c.c.}{=} \mathrm{P}\left(X \geq 15.5\right)$ $= 0.0172875127 \approx 0.0173$	